



Faculty Workload and Timetable Management System

DBMSProject Presentation

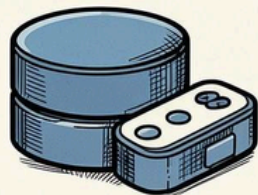
Presented by: Lingampally Mohith — Roll No: 25WU0102137

Guided By : Dr. Kiran Mayee Adavala

Catabare Deooo							
				1	(1)	(1)	
				2			
					2		

Problem Statement

- Departments must allocate courses, compute faculty workload, and produce conflict-free timetables
- Manual/spreadsheet planning causes: faculty, room, section clashes
- Difficulty calculating workload, handling substitutions and timetable changes
- Solution: centralized relational database to store and enforce scheduling rules



- ☒ Objessiinciues
- ☒ Rish. Daatiacar
- ☒ Abstart Flobits
- ☒ Draecticirors
- ☒ Motvcan Ceseon
- ☒ Gleaint Fegecronds
- ☒ Dempiute Xbleen
- ☒ Objecries:

Objectives

- Manage departments, programmes, courses, sections, faculty
- Allocate courses to faculty; calculate workload
- Create conflict-free timetables using rooms & time slots
- Detect clashes; enforce workload limits and room capacity
- Manage substitutions; generate reports

Users & Functional Requirements

Users

- 1. Department / Timetable Administrator
- 2. Faculty

Admin Functions

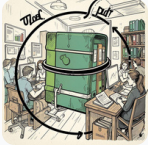
- Maintain master data (faculty, courses, rooms, slots)
- Allocate courses; prepare timetable; check clashes; generate reports

Faculty Functions

- View assignments, workload, timetable, substitution info



Major Entities



Department

dept_id,
dept_name



Programme

programme_id,
dept_id



Course

course_id,
programme_id,
credits



Section

section_id,
course_id,
strength



Faculty

faculty_id, dept_id



Room / Time Slot

room_id, capacity
• slot_id, day,
start_time,
end_time



Assignment

assignment_id,
faculty_id,
section_id



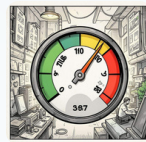
Timetable Entry

entry_id,
assignment_id,
room_id, slot_id



Substitution

sub_id,
original_faculty_id
,
substitute_facul t y
_id, entry_id

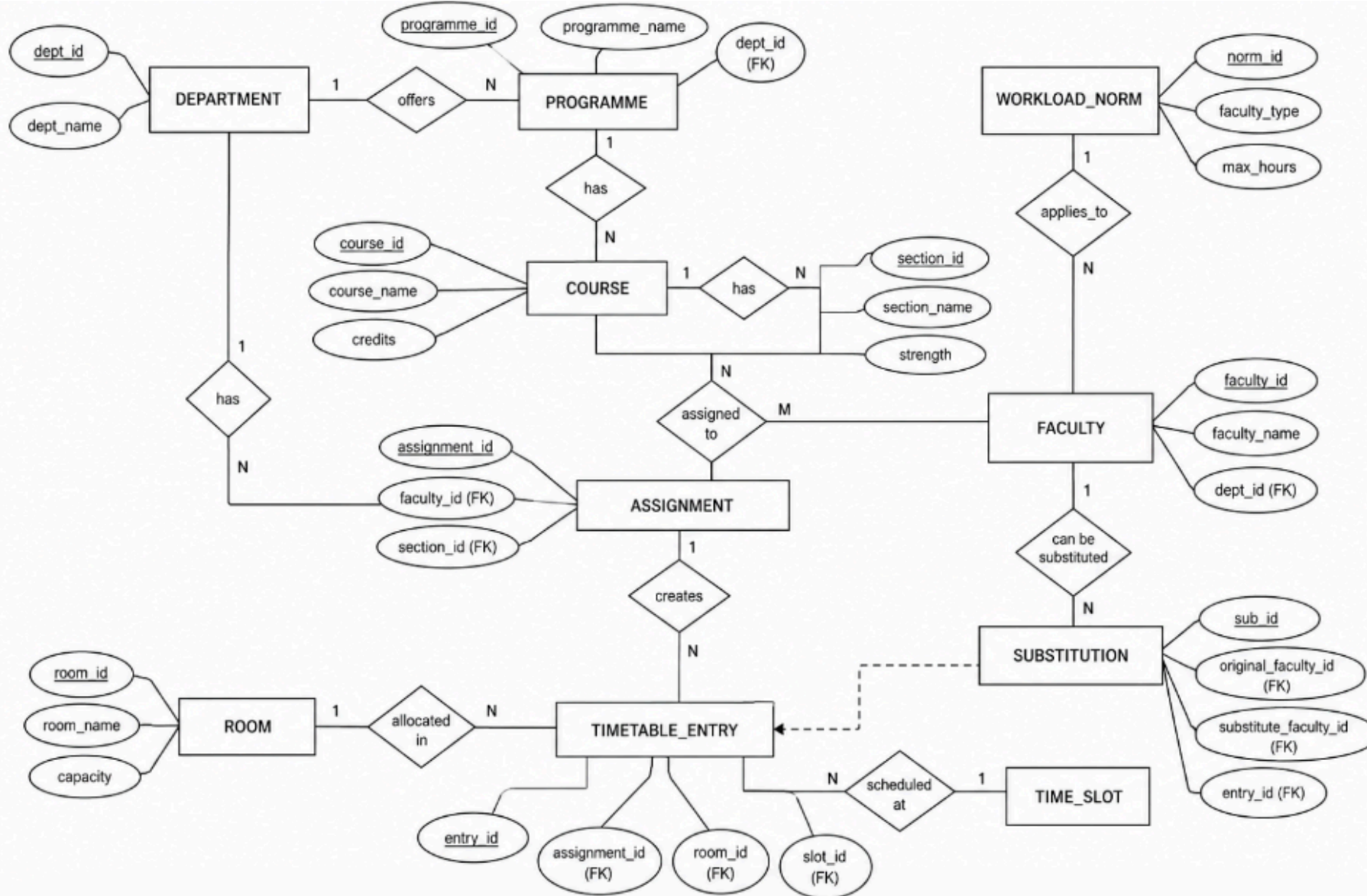


Workload Norm

norm_id,
faculty_type,
max_hours

ER Diagram

PKs shown on each entity; FK links labelled (dept_id, programme_id, course_id, faculty_id, assignment_id, room_id, slot_id)



Relational Schema (PK / FK)

DEPARTMENT(dept_id PK, dept_name)

PROGRAMME(programme_id PK, dept_id FK, programme_name)

COURSE(course_id PK, programme_id FK, course_name, credits)

SECTION(section_id PK, course_id FK, section_name, strength)

FACULTY(faculty_id PK, dept_id FK, faculty_name)

WORKLOAD_NORM(norm_id PK, faculty_type, max_hours)

ROOM(room_id PK, room_name, capacity)

TIME_SLOT(slot_id PK, day, start_time, end_time)

ASSIGNMENT(assignment_id PK, faculty_id FK, section_id FK)

TIMETABLE_ENTRY(entry_id PK, assignment_id FK, room_id FK, slot_id FK)

SUBSTITUTION(sub_id PK, original_faculty_id FK, substitute_faculty_id FK, entry_id FK)

Notation: PK = Primary Key, FK = Foreign Key

Business Rules & Constraints

- No overlapping faculty timetable entries (CHECK / unique scheduling logic)
- No overlapping room allocation
- No overlapping section allocation
- Faculty workload $\leq$ Workload_Norm.max_hours
- Room.capacity $\geq$ Section.strength

Key Constraints

- PRIMARY KEY — uniqueness per table
- FOREIGN KEY — referential integrity between entities
- UNIQUE — prevent duplicate assignments or entries
- NOT NULL — required attributes (ids, times)
- CHECK — enforce ranges (e.g., start_time < end_time)
- DEFAULT — sensible defaults for status/created_at

These constraints maintain data integrity and prevent invalid schedules.

Sample SQL Queries

Query 1 — Faculty workload

```
SELECT f.faculty_name,  
       COUNT(a.assignment_id) AS assigned_classes  
FROM Faculty f  
JOIN Assignment a ON f.faculty_id = a.faculty_id  
GROUP BY f.faculty_id, f.faculty_name;
```

Uses JOIN to combine tables; GROUP BY aggregates per faculty

Query 2 — Timetable

```
SELECT f.faculty_name, s.section_name,  
       r.room_name, t.day, t.start_time, t.end_time  
FROM Timetable_Entry t  
JOIN Assignment a ON t.assignment_id =  
a.assignment_id  
JOIN Faculty f ON a.faculty_id = f.faculty_id  
JOIN Section s ON a.section_id = s.section_id  
JOIN Room r ON t.room_id = r.room_id;
```

JOIN chain retrieves readable timetable rows

outcomes:



Efficient Faculty Workload Management

Calculate and monitor the teaching workload of each faculty member.



Conflict-Free Timetable Generation

Reduce faculty, room and section clashes during timetable preparation



Centralized Database Management

Store faculty, courses, sections, rooms, time slots and timetable information in one database.



Better Course Allocation

Assign courses/sections to appropriate faculty members while considering workload limits.



Improved Resource Utilization

Efficiently manage classrooms and available time slots based on capacity and availability.



Easy Reporting and Decision Making

Generate reports such as faculty workload, room utilization, timetable, clashes and substitutions.